

FRAGMENTED ID RESOLUTION USING DEEP LEARNING

The Erdos Institute Spring 2026 Bootcamp

Noimot Bakare Ayoub,
Pedro Fontanarossa,
Arpith Shanbhag,
Dharineesh Somisetty



Motivation

Problem: Identity data in real-world systems are noisy, and fragmented. Small inconsistencies can cause a single person to appear as multiple records, creating major risks, ranging from:

Stakeholder:



Financial Services:
Vulnerability to **synthetic identity fraud** and account takeover.



Healthcare:
Patient mismatches, duplicated charts, and potential clinical risk.



Government & Public Sector:
Duplicate citizen records, complicating benefits, taxation, and verification.

Task: Traditional matching methods break down in these high-noise environments, motivating the need for a more robust approach.

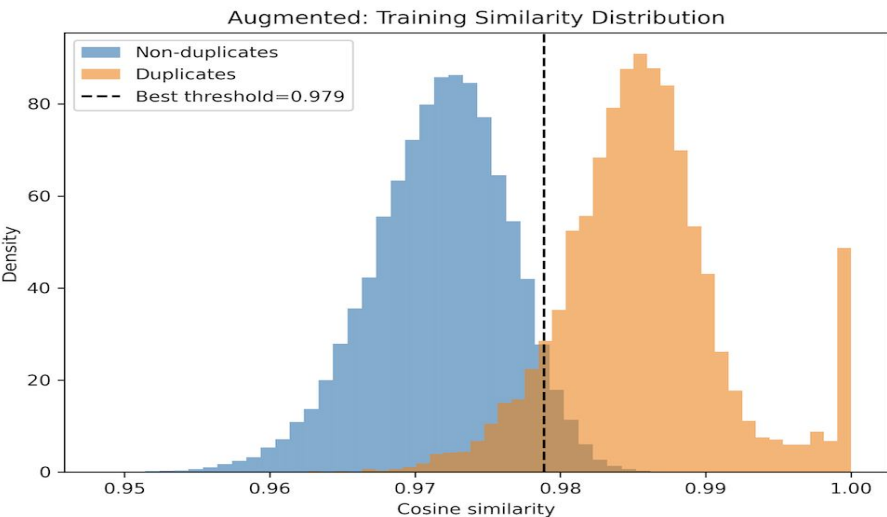
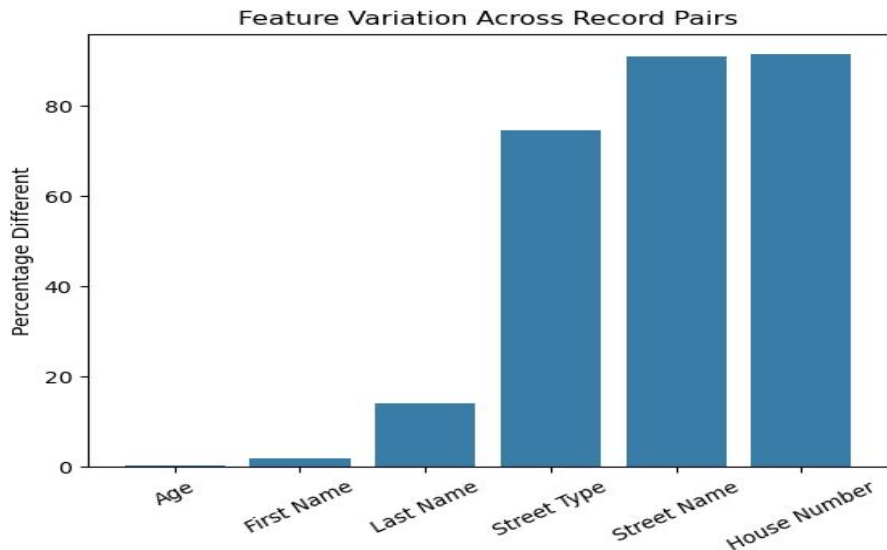
 Elisabeth Millar 123 Main St. Raleigh, NC 27601 Typeo	 Liz Miller-Davis 123 Main St. Raleigh, NC 27601 Nickname	 Elizabeth M. Miller 123 Main St. Raleigh, NC 27601 Different Last Name	 Elizabeth Miller 123 Main Sc, 27602 Typos
--	---	---	--



Why Simple Thresholding Fails:

1. Duplicates vary across fields

→ Same pair can be “close” in one attribute and “far” in another



2. Similarity scores overlap

→ Thresholds misclassify borderline cases

Identity Matching For Rule Based Learning

CNN Embedding + Cosine Similarity Baseline

 **Person A — Same Individual (Fragmented)**

(Positive Pairs)



Elisabeth Millar
123 Main St,
Raleigh, NC 27601

Typo



Liz Miller
123 Main St,
Raleigh, NC 27601

Nickname



Elizabeth Miller-Davis
123 Main St,
Raleigh, NC 27601

Different Last Name



Elizabeth Miller-Davis
123 Main St,
Raleigh, NC 27602

ZIP Variation



Person B — Different Individuals (Similar Identities)

(Hard Negatives)



Emily Carter
123 Main St,
Raleigh, NC 27601

Nickname



Emma Carter
123 Main St,
Raleigh, NC 27601

Shortened Name



Ava Thompson
123 Main St,
Raleigh, NC 27601

ZIP Variation



Chloe Anderson
123 Main St,
Raleigh, NC 27603

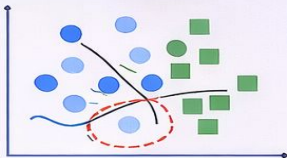
ZIP Variation



Hard Case: Different last names + similar attributes → High confusion risk

Manually Engineered Features

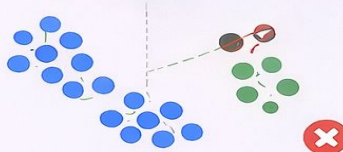
Model 1: Logistic Regression + XGBoost



- ✓ Solves typos, nicknames, ZIP variance
- ✗ Struggles with last name differences

Embedding Only

Model 2: Embedding + Cosine Similarity

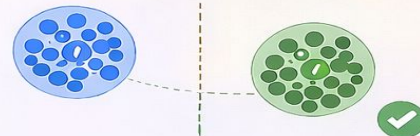


- ✓ Captured some variations
- ✗ Poor separation on hard cases



Embedding + Weights + Siamese

Model 3: Siamese Network

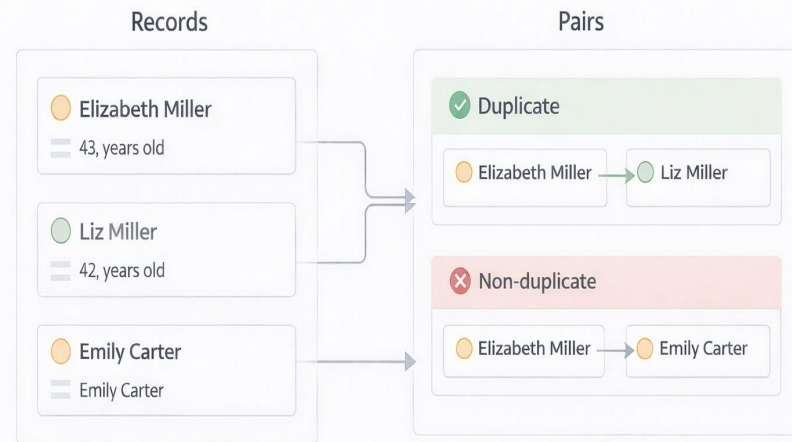


- ✓ Resolving fragmentation across variations
- ✓ Distinguishes similar but different individuals

Source: Hasso Plattner Institute (HPI)

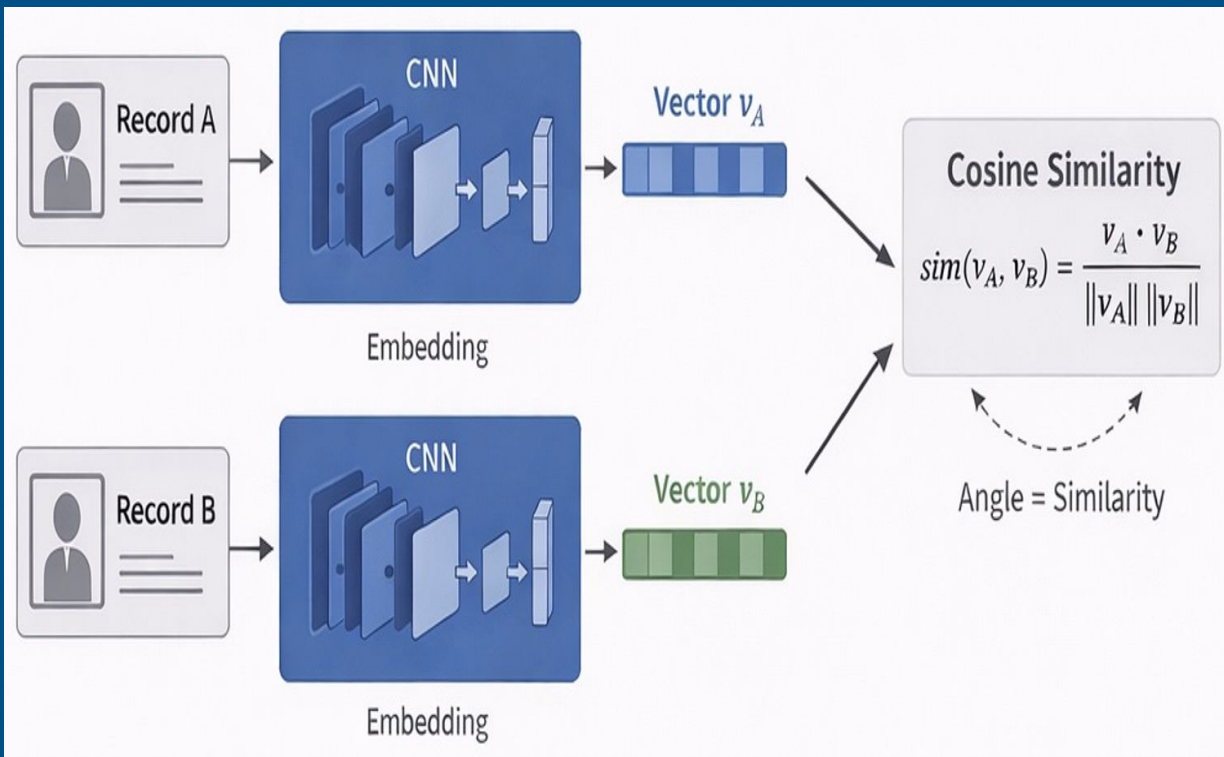
North Carolina State Board of Election (NCSBE)

- **Voter records:** 14,183
- **Labeled pairs**
 - 9,891 **duplicate**
 - 98,142 **non-duplicate**
- Augmented data to reflect real-world variations
- Constructed hard negatives to capture confusable identities
- Group based entity-disjoint train / validation / test split for evaluation



Learning Identity Similarity

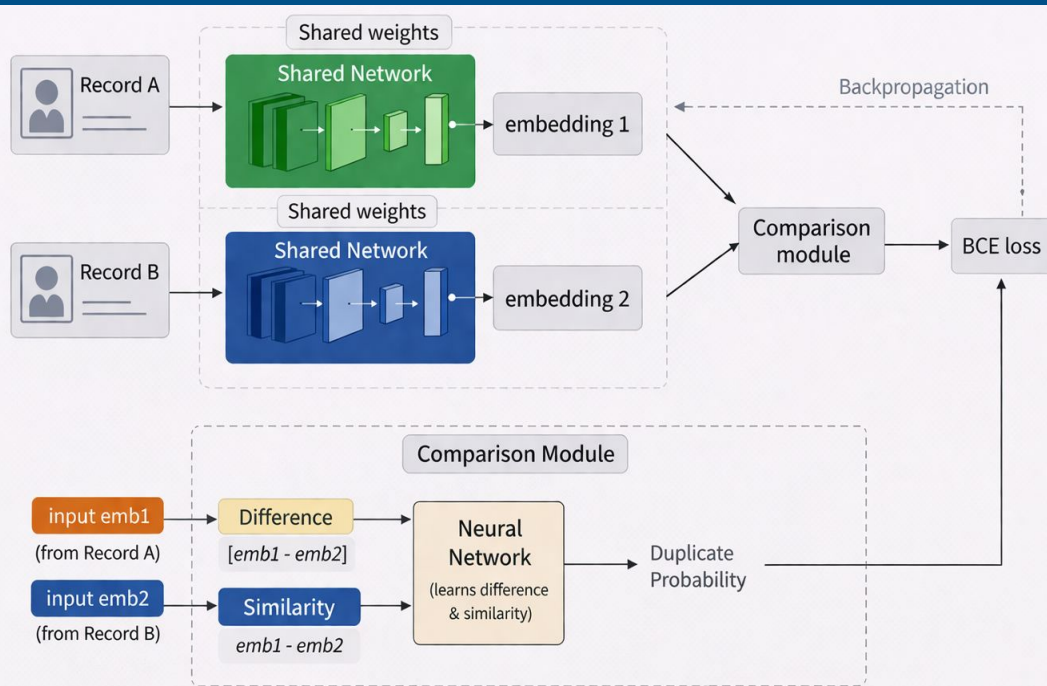
CNN Embedding + Cosine Similarity Baseline



- Converts records into embeddings
- Uses cosine similarity to compare record pairs
- Treats all differences equally, struggles on complex cases

Siamese Network

Learning What Matters for Identity Matching



- Siamese network encodes both records
- Learns similarity from how records agree and differ
- **Adaptive similarity function improves hard case detection**
- **Assigns higher weight to meaningful differences**

Results

Siamese model significantly improves performance on hard cases

Model	Precision		Recall		F1	
	Overall	Hard	Overall	Hard	Overall	Hard
XGBoost	.97	.86	.97	.82	.97	.84
Embedding + Cosine (DL Baseline)	.94	.65	.91	.78	.92	.71
Siamese + Weights	.99	.93	.99	.99	.99	.96



25 pts
(Hard F1: 0.71 → 0.96)

Overall – all data
Hard – hard case subset

LinkID App – Identity Matching Simplified

IDENTITY MATCHING

LinkID

Detects and ranks duplicate identities across messy datasets for accurate, scalable identity resolution.

1. Load a voter-registration dataset by path or file upload.
2. Choose whether you want a full duplicate scan or a single-record check.
3. Review the summary, confirm strong duplicates, and handle human-review cases when needed.
4. Export duplicate and review queues for downstream action.

[Technical API docs](#)

STEP 1

Load a Dataset

Use File Path

Upload File

Dataset path on this machine

data/processed/ncvoters_prepared.tsv

Load Dataset

Use forward slashes on Windows to avoid escaping issues.

Dataset Status

Clear Dataset

Source

data/processed/ncvoters_prepared.tsv

Records

14183

Detects and ranks duplicate voter registration records for accurate identity resolution.

Step 1:

Upload voter registration data

Step 2:

Detect and rank likely duplicates

Step 3:

Review high confidence matches and cases close to the threshold.

Step 4:

Export

Business KPI Impact Summary



- Normal Review: 5 min
- Hard Review: 20 min
- Analyst cost: \$40/hr

-
- Average review cost: \$3.33



Manual review reduction:

31.85%

Total cases: 7,471

Reviews avoided: 2,381



~ \$7,930 saved

(2,381 x \$3.33)

Our model eliminates nearly 1 in 3 manual reviews

Cost savings + faster processing



Acknowledgments



Lindsay Warrenbug

Marcos Ortiz



Roman Holowinsky



Hasso Plattner Institute

About Us

Noimot Bakare Ayoub

Data Scientist, PhD in Economics | Specializing in Causal Inference and Predictive Modeling

<https://www.linkedin.com/in/noimot-bakare-ayoub/>

Pedro Fontanarossa

Data Scientist + ML Engineer, PhD in Computational Biology | Specializing in Time Series Analysis

<https://www.linkedin.com/in/pedro-fontanarrosa/>

Arpith Shanbhag

Quantitative Researcher, PhD in Mathematics | Specializing in Machine Learning for Financial Markets

<https://www.linkedin.com/in/arpithshanbhag/>

Dharineesh Somisetty

Data Scientist, Masters in Computational and Applied Math | Specializing in Theoretical Machine Learning

<https://www.linkedin.com/in/dharineesh-somisetty/>